

MANAGEMENT REVIEW

Management without theatre.

PEOPLE

SHIFT WORK AND NIGHT WORK

Who works while others sleep, how risk climbs night after night, a roster that respects the body clock, what “probably carcinogenic” means, what EU law counts, and a card for checking a roster.



+36%

higher risk of injuries or accidents on the fourth night shift in a row than on the first, pooled from seven published studies.

FOLKARD ET AL., 2005

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NO. 45 · PEOPLE

In this issue

Hotels, hospitals, warehouses and transport run through the night, and someone draws up the roster. This issue looks at what the evidence says about working at night, what the law requires, and which choices in a roster are in a manager's hands.

In a 2024 European survey, 21% of workers worked shifts and 17% worked at night at least once a month. Pooled studies show the risk of incidents rising over successive night shifts and with shifts longer than eight hours. The UK Health and Safety Executive advises forward, fast-rotating rosters with few nights in a row, though the evidence on rotation direction is very uncertain. IARC classifies night shift work as probably carcinogenic to humans, a statement about the hazard, not about the size of the risk. EU law sets limits and a free health assessment for night workers.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure

Every figure has its source and year at the foot of its page.

Our reading

Where the editors interpret rather than the research, it says so.

Practice

The steps and the card are proposals to try, not research results.

COVER STORY

When the others sleep

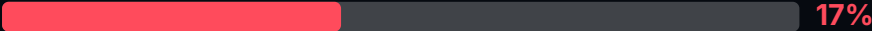
For the European Working Conditions Survey 2024, Eurofound interviewed 36,644 people face to face in 35 countries, Albania among them. Its highlights on working time:

WORKERS IN EUROPE, 2024

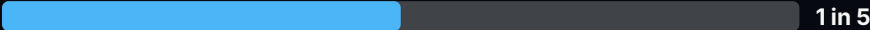
Worked shifts



Worked 2+ hours between 22:00 and 05:00, at least monthly



Under 11 hours' rest between working days, at least monthly



The 2021 telephone survey asked differently: in the EU, 21% worked at night sometimes or more often, 25% of men and 17% of women, most often in security jobs and in care, such as medical staff.

OUR READING

Night work is not a niche. In most operations that never close, someone on the team works while the manager sleeps.

Sources: Eurofound, 2026; Eurofound, 2022 (via EU-OSHA, OSHwiki)

Self-reports. By Eurofound's account, night and weekend work have fallen since 2015 while shift work has stayed stable; 2021 asked differently. No separate figure for Albania in the highlights.

More in the essay

What a Night Auditor learns about a business that most managers never see

THE NUMBERS

Night after night

Simon Folkard, David Lombardi and Philip Tucker pooled published studies of injuries and accidents at work: seven counted them over at least four night shifts in a row, five of these also over morning or day shifts.

HIGHER RISK THAN ON THE FIRST SHIFT OF A RUN, 2005

■ Morning or day shifts

■ Night shifts

2nd shift in a row

■ +2%

■ +6%

3rd shift in a row

■ +7%

■ +17%

4th shift in a row

■ +17%

■ +36%

+30%

on the night shift against the morning shift; afternoon: +18%

+27%

on 12-hour shifts against 8-hour shifts; 10 hours: +13%

OUR READING

The fourth night costs more than the first. How many nights in a row is decided in the roster.

Source: Simon Folkard, David A. Lombardi & Philip T. Tucker, Industrial Health, 2005

Relative risks pooled from a few studies, not the risk of one workplace. Shift length, nights in a row and breaks are to be judged together, the authors say.

THE MODEL

A roster that respects the night

The UK Health and Safety Executive's guide Managing shiftwork (HSG256, 2006) turns the research into advice for those who design rosters. Its main points:

01

Forward

morning, afternoon, then night; limited evidence that the body clock adapts faster

02

Fast or slow

change every 2–3 days or every 3–4 weeks, not every week or two

03

Few in a row

nights, long shifts, early starts: 2–3 in a row, then 2–3 days off

04

Length

at most 12 hours with overtime; 8 for nights and safety-critical work

05

Rest

11 hours between shifts; two nights' sleep when switching between days and nights

06

Timing

no starts before 7:00 unless needed; hard tasks away from low alertness

How strong is the evidence? In a 2023 Cochrane review of 11 studies, forward and faster rotation may reduce sleepiness on shift and may not change sleep quality, but the evidence is very uncertain: on direction, one trial with 62 participants gave usable data.

OUR READING

Forward rotation is sensible advice, not a proven cure. The previous page points more clearly at nights in a row and shift length.

Sources: Health and Safety Executive, 2006; Hulsegge et al., Cochrane Database of Systematic Reviews, 2023 (via PubMed)

The advice is HSE's, written for Britain; the summary in the cards is the editors'. The review is Hulsegge et al. (2023).

WHAT THE RESEARCH SAYS

Probably carcinogenic

In June 2019, 27 experts from 16 countries met at the International Agency for Research on Cancer (IARC) in Lyon and classified night shift work, work during the usual sleeping hours of the population, as probably carcinogenic to humans: Group 2A.

01**In humans**

limited evidence:
positive associations
with cancers of the
breast, prostate, colon
and rectum; chance,
bias or confounding not
ruled out

02**In animals**

sufficient evidence for
altering the light–dark
schedule

03**Mechanisms**

strong evidence in
experimental systems:
immunosuppression,
chronic inflammation,
cell proliferation

The class says how strong the evidence of a hazard is, not how large the risk is. Working as a hairdresser or barber is in the same group, IARC notes, and agents in one group can carry very different risks. In 2007 an earlier working group had reached the same class for shift work that involves circadian disruption.

OUR READING

For a manager the lesson is not a diagnosis but a duty of care: plan night work as a recognised hazard, and leave health questions to doctors.

Sources: IARC, 2020; IARC, 2019

IARC makes no recommendations for individuals; it tells those who are concerned to consult their physician. This issue gives no medical advice.

HOW IT IS MEASURED

What the law counts

The EU Working Time Directive 2003/88/EC sets minimum rules; Articles 8 to 13 cover night work. Night time is at least seven hours, always including midnight to 05:00; a night worker normally works at least three hours of the working day in it.

01	Art. 8, length	on average at most 8 hours in 24; with special hazards or heavy strain, at most 8 in any 24 hours with night work
02	Art. 9, health	a free, confidential health assessment before assignment and at regular intervals; suitable day work where possible for health problems linked to night work
03	Art. 12, protection	protection that fits the work, with services available at all times
04	Art. 13, pattern	adapting work to the worker, easing monotonous work and work at a fixed pace

HYPOTHETICAL EXAMPLE, FOUR WEEKS OF A NIGHT TEAM

Nights in a row	longest run 5; 2 runs longer than 3
Rest	3 changeovers with under 11 hours
Health	assessment offered to 6 of 8 night workers

Three counts to check a roster against every month. The numbers are invented.

Source: European Parliament and Council of the EU, 2003 (via legislation.gov.uk)

Minimum rules; national law and collective agreements set the details and may go further. The 11 hours of daily rest are in Article 3. Not legal advice.

TOOL OF THE ISSUE

The roster check card

One roster period, one team. Fill it in before the roster is published, and again after it has been worked: the plan and what really happened can differ.

01 Period and team dates, team, how many people work nights

02 Nights in a row the longest run, and the rest days after it

03 Longest shift hours with overtime; which nights are safety-critical

04 Shortest rest from the end of one shift to the start of the next; any under 11 hours?

05 Rotation forward or backward; every few days or every few weeks

06 Health and handover health assessment offered; what the night hands to the morning, and to whom

Sources: Simon Folkard, David A. Lombardi & Philip T. Tucker, *Industrial Health*, 2005; Health and Safety Executive, 2006; European Parliament and Council of the EU, 2003 (via legislation.gov.uk)

A practice proposed by the editors, after Folkard et al. (2005), HSE (2006) and Directive 2003/88/EC. It does not replace legal or medical advice.

Open the tool: [Shift Handover](#)

stivencatalyst.com/tools/shift-handover/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

European Working Conditions Survey 2024: Highlights

Eurofound, 2026

<https://eurofound.shorthandstories.com/european-working-conditions-survey-2024-highlights/index.html>

Working conditions in the time of COVID-19: Implications for the future

Eurofound, 2022 · via EU-OSHA, OSHwiki

<https://www.eurofound.europa.eu/publications/report/2022/working-conditions-in-the-time-of-covid-19-implications-for-the-future>

Shiftwork: Safety, Sleepiness and Sleep

Simon Folkard, David A. Lombardi & Philip T. Tucker, Industrial Health, 2005

https://www.jstage.jst.go.jp/article/indhealth/43/1/43_1_20/_article

Managing shiftwork: Health and safety guidance (HSG256)

Health and Safety Executive, 2006

<https://www.hse.gov.uk/pubns/priced/hsg256.pdf>

Adapting shift work schedules for sleep quality, sleep duration, and sleepiness in shift workers

Hulseegge et al., Cochrane Database of Systematic Reviews, 2023 · via PubMed

<https://doi.org/10.1002/14651858.CD010639.pub2>

Night Shift Work (IARC Monographs, Volume 124)

IARC, 2020

<https://publications.iarc.who.int/Book-And-Report-Series/Iarc-Monographs-On-The-Identification-Of-Carcinogenic-Hazards-To-Humans/Night-Shift-Work-2020>

IARC Monographs Meeting 124: Night Shift Work. Questions and Answers

IARC, 2019

https://www.iarc.who.int/wp-content/uploads/2019/07/QA_Monographs_Volume124.pdf

Directive 2003/88/EC concerning certain aspects of the organisation of working time

European Parliament and Council of the EU, 2003 · via legislation.gov.uk

<https://eur-lex.europa.eu/eli/dir/2003/88/oj/eng>

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked “via”. The editors' interpretation is marked “Our reading”. Figures that could not be confirmed are not in the issue.



MANAGEMENT REVIEW

Management without theatre.

Every issue, one management question, checked against the best research.

ALL ISSUES

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